VECTOR

Vector Legacy Converter

Technical Reference

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Authors	Thomas Ziegler
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Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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1 Introduction

The Vector Legacy Converter (VLC) supports the migration of legacy embedded software to the AUTOSAR software architecture. The VLC is a console application which transforms one or more DBC-, LDF- and Fibex files into an AUTOSAR System Description and its ECU Extracts. The VLC is typically called by the DaVinci Project Assistant (DPA), but it can also be used as a stand-alone tool. The resulting ECU Extracts will serve as input for the Initial EcuC Generator.

1.1 Supported AUTOSAR versions

The Vector Legacy Converter supports several AUTOSAR versions selectable via start parameter -r as shown in the following table.

-r option	AUTOSAR version
-r 31	AUTOSAR 3.1.4
-r 32	AUTOSAR 3.2.1
-r 40	AUTOSAR 4.0.3
-r 41	AUTOSAR 4.1.2
-r 42	AUTOSAR 4.2.1
-r 00042	AUTOSAR 4.2.1
-r 00043	AUTOSAR 4.3.0
-r 00046	AUTOSAR 4.4.0
-r 00048	AUTOSAR 4.5.0

2 Functional Description

The VLC analyses legacy communication databases, and it maps their communication elements to AUTOSAR System Description elements. There are no standards or established rules which define such a mapping between legacy communication databases and System Descriptions. For this reason, the VLC defines its own AUTOSAR mapping rules which aim at preserving the semantics of the original communication databases. These generic rules may be supplemented with OEM-specific rules.

The AUTOSAR System Description allows various modeling variants w.r.t. to, e.g., naming conventions and package structures. The VLC imposes fixed modeling rules which define a common namespace for DBC-, LDF- and Fibex transformations. The VLC modeling rules are not coordinated with the AUTOSAR transformations of other tools from other vendors. The transformation results of the VLC and other tools may appear rather different.

The VLC supports no user interaction, and thus the transformation between legacy formats and AUTOSAR System Descriptions is always the same. However, the VLC still identifies the manufacturer of a communication database, and it applies OEM-specific rules. These OEM-specific rules must be implemented in advance.

The calling conventions and options of the VLC are specified with the following help text:

```
Usage: LegacyDb2SystemDescrConverter [options] <file|dir> [<file|dir> ...]
[extfile]

Create an AUTOSAR System Description out of one or more DBC, LDF or Fibex
communication databases.

Options:
  -h, --help                          Show this help
  -a, --adoptname                     Adopt DBC filename as cluster name
  -e, --extract                       Create ECU extracts
  -r, --release <31|32|40|41|42
                |00042|00043|00046
                |00048>               AUTOSAR release
  -s, --showAsrVersions               Display supported AUTOSAR schema version
  -v, --version                       MSR version
  -p, --ports                         Create port interface catalog
  -o, --output <file|dir>             Output file or directory

Parameters:
  <file|dir>                          Input file (*.dbc, *.ldf, *.xml) or directory
  extfile                             Extension file (*.vsde)
```

Table 2-1 Vector Legacy Converter help text.

The options may also be specified in a file called LegacyDb2SystemDescrConverter.config which resides in the same directory as the exe file LegacyDb2SystemDescrConverter.exe. In this way, options can be defined which are processed, e.g., when calling the VLC from the DPA.

With the option ‘-r’ or ‘--release’ the AUTOSAR schema version can be selected. Currently the AUTOSAR schema versions listed in chapter 1.1 are supported. The VLC implements

those schema versions which are required by the Vector tool chain, especially by the Initial EcuC Generator. The VLC does not aim at supporting arbitrary AUTOSAR shema versions.

Please note that the DBC-, LDF- and Fibex transformations are rather sophisticated, and thus we can only provide a general survey with this document. To identify the AUTOSAR mapping more in detail, the user may, e.g., perform minor changes to a communication database, and then compare the transformation results before and after these changes. The VLC defines a fixed order for all AUTOSAR elements, so two System Descriptions can be easily diffed.

2.1 DBC Transformation

The DBC file format is based on a network-specific object model and on user-defined attributes. The former can be transformed in a generic way to AUTOSAR, while the latter often require an OEM-specific transformation. The table below shows how CAN network objects are mapped to AUTOSAR elements of the selected version as described in chapter 1.1.

CAN network object	AUTOSAR element
Signal	SystemSignal ShortName = Signal.Name Length = Signal.Bitcount BooleanType|IntegerType|RealType ShortName = "DT_" + Signal.Name LowerLimit = (Signal.Min-Signal.Offset)/Signal.Factor UpperLimit = (Signal.Max-Signal.Offset)/Signal.Factor CompuMethod ShortName = "CM_" + Signal.Name Unit ShortName = "U_" + Signal.Unit CompuInternalToPhys.CompuScale LowerLimit = Signal.TextualEncoding.LowerBound UpperLimit = Signal.TextualEncoding.UpperBound CompuConst = "Cx<Limit>_" + Signal.TextalEncoding.Text CompuInternalToPhys.CompuScale.CompuRationalCoeffs CompuNumerator = Signal.Offset, Signal.Factor
SignalGroup	SystemSignalGroup ShortName = "SG_" + SignalGroup.Name
CANBus	CanCluster ShortName = CANBus.Attributes.DBName ProtocolName = "CAN" PhysicalChannel ShortName = "CHNL"
CANBus .CANFrame	Frame ShortName = CANFrame.Name + "__" + CanCluster.ShortName FrameLength = CANFrame.DLC SignalIPdu|MultiplexedIPdu|DcmIPdu|NmPdu|NPdu ShortName = CANFrame.Name + "__" + CanCluster.ShortName Length = 8*CANFrame.DLC Frame.PduToFrameMapping ShortName = CANFrame.Name PackingByteOrder = Intel StartPosition = 0 PhysicalChannel.CanFrameTriggering ShortName = "FT_" + CANFrame.Name Identifier = CANFrame.ID PhysicalChannel.IPduTriggering ShortName = "PT_" + CANFrame.Name
CANBus .CANFrame .MappedSignal .Signal	ISignal ShortName = Signal.Name + "__" + CANFrame.Name + "__" + CanCluster.ShortName SignalIPdu.ISignalToIPduMapping ShortName = Signal.Name PackingByteOrder = MappedSignal.Intel|Motorola StartPosition = MappedSignal.Startbit PhysicalChannel.SignalTriggering ShortName = "ST_" + Signal.Name + "__" + CANFrame.Name
CANBus .CANFrame .MappedMultiplexorSignal .MultiplexorSignal	MultiplexedIPdu.SelectorField ByteOrder = MappedMultiplexorSignal.Intel|Motorola Length = MultiplexorSignal.Bitcount StartPosition = MappedMultiplexorSignal.Startbit
CANBus .CANFrame .MappedMultiplexorSignal .MappedMultiplexedSignal .MultiplexedSignal	MultiplexedIPdu.DynamicPart.DynamicPartAlternative SelectorFieldCode = MappedMultiplexedSignal.MultiplexorValue SignalIPdu ShortName = CANFrame.Name + "_Mx<Code>__" + CanCluster.ShortName Length = 8*CANFrame.DLC PhysicalChannel.IPduTriggering

	ShortName = "PT_" + CANFrame.Name + "_Mx<Code>"
CANNode	EcuInstance ShortName = CANNode.Name ComProcessingPeriod = 0.001 EcuInstance.CanCommunicationController ShortName = "CT_" + CanCluster.ShortName EcuInstance.CommunicationConnector ShortName = "CN_" + CanCluster.ShortName EcuInstance.AssociatedIPduGroup(Rx) ShortName = CANNode.Name + "__" + CanCluster.ShortName + "_Rx" EcuInstance.AssociatedIPduGroup(Tx) ShortName = CANNode.Name + "__" + CanCluster.ShortName + "_Tx"
CANNode .RxCANFrame	CommunicationConnector.FramePort ShortName = "FP_" + RxCANFrame.Name + "_Rx" Direction = In CommunicationConnector.IPduPort ShortName = "PP_" + RxCANFrame.Name + "_Rx" Direction = In
CANNode .RxCANFrame .MappedSignal .Signal	CommunicationConnector.SignalPort ShortName = "SP_" + Signal.Name + "__" + RxCANFrame.Name + "_Rx" Direction = In
CANNode .TxCANFrame	CommunicationConnector.FramePort ShortName = "FP_" + TxCANFrame.Name + "_Tx" Direction = Out CommunicationConnector.IPduPort ShortName = "PP_" + TxCANFrame.Name + "_Tx" Direction = Out
CANNode .TxCANFrame .MappedSignal .Signal	CommunicationConnector.SignalPort ShortName = "SP_" + Signal.Name + "__" + TxCANFrame.Name + "_Tx" Direction = Out

Table 2-2 Transformation of CAN network objects.

The subsequent table shows how user-defined attributes are processed by the AUTOSAR transformation.

Attribute	Autosar element
Signal.GenSigStartValue	ConstantSpecification ShortName = "C_" + Signal.Name BooleanLiteral|IntegerLiteral|RealLiteral ShortName = "C_" + Signal.Name Value = Signal.GenSigStartValue
CANFrame.GenMsgILSupport	SignalIPdu|MultiplexedIPdu
CANFrame.DiagRequest CANFrame.DiagResponse CANFrame.DiagUUDTResponse CANFrame.DiagState	DcmIPdu NPdu
CANFrame.NmAsrMessage	NmPdu
CANFrame.GenMsgSendType	SignalIPdu.IPduTimingSpecification
CANFrame.GenMsgCycleTime CANFrame.GenMsgStartDelayTime CANFrame.GenMsgNrOfRepetition CANFrame.GenMsgCycleTimeFast	SignalIPdu.IPduTimingSpecification.CyclicTiming RepeatingTime = CANFrame.GenMsgCycleTime|GenMsgCycleTimeFast StartingTime = CANFrame.GenMsgStartDelayTime SignalIPdu.IPduTimingSpecification.EventControlledTiming NumberOfRepeats = CANFrame.GenMsgNrOfRepetition RepetitionPeriod = CANFrame.GenMsgCycleTimeFast
CANFrame.GenMsgDelayTime	SignalIPdu.IPduTimingSpecification MinimumDelay = CANFrame.GenMsgDelayTime
Signal.GenSigInactiveValue	SignalIPdu.IPduTimingSpecification.TransmissionModeCondition MaskedNewDiffersX.X = Signal.GenSigInactiveValue
Signal.GenSigSendType	SignalIPdu.ISignalToIPduMapping TransferProperty = Pending|Triggered|TriggeredWithoutRepetition |TriggeredOnChange|TriggeredOnChangeWithoutRepetition
Signal.GenSigTimeoutTime	EcuInstance.CommunicationConnector.SignalPort Timeout = Signal.GenSigTimeoutTime
CANFrame.DiagConnection CANFrame.CanTpBs CANFrame.CanTpSTmin	CanCluster.PhysicalChannel.CanTpConnectionChannel DataPdu.DiagConnection = FlowControlPdu.DiagConnection BlockSize = CANFrame.CanTpBs MinimumSeparationTime = CANFrame.CanTpSTmin
CANBus.DBName	CanCluster ShortName = CANBus.DBName
CANBus.Baudrate	CanCluster Speed = CANBus.Baudrate
CANBus.BaudrateCANFD	AbstractCanCluster canFdBaudrate = CANBus.BaudrateCANFD
CANBus.NBTMin CANBus.SamplePointMin CANBus.SyncJumpWidthMin CANBus.NBTMax CANBus.SamplePointMax CANBus.SyncJumpWidthMax	EcuInstance.CanCommunicationController.ConfigurationRequirements MinNumberOfTimeQuantaPerBit = CANBus.NBTMin MinSamplePoint = CANBus.SamplePointMin MinSyncJumpWidth = CANBus.SyncJumpWidthMin MaxNumberOfTimeQuantaPerBit = CANBus.NBTMax MaxSamplePoint = CANBus.SamplePointMax MaxSyncJumpWidth = CANBus.SyncJumpWidthMax
CANBus.NmAsrBaseAddress CANBus.NmAsrMessageCount CANBus.NmAsrCanMsgCycleTime	CanCluster NmLowerCanID = CANBus.NmAsrBaseAddress NmUpperCanID = CANBus.NmAsrBaseAddress + CANBus.NmAsrMessageCount-1 CanCluster.AdminData.CanNmConfiguration CanNmBaseAddress = CANBus.NmAsrBaseAddress CanNmMessageCount = CANBus.NmAsrMessageCount CanNmMsgCycleTime = CANBus.NmAsrCanMsgCycleTime
CANBus.NmAsrRepeatMessageTime CANBus.NmAsrTimeoutTime	CanCluster NmRepeatMessageStateTime = CANBus.NmAsrRepeatMessageTime NmTimeoutTime = CANBus.NmAsrTimeoutTime

CANBus.NmAsrWaitBusSleepTime	NmWaitBusSleepTime = CANBus.NmAsrWaitBusSleepTime
CANNode.NmAsrNodeIdentifer	EcuInstance.CommunicationConnector NmAddress = CANNode.NmAsrNodeIdentifer
CANNode.NmAsrCanMsgCycleOffset CANNode.NmAsrCanMsgReducedTime	EcuInstance.AdminData.CanNmConfiguration CanNmMsgCycleOffset = CANNode.NmAsrCanMsgCycleOffset CanNmMsgReducedTime = CANNode.NmAsrCanMsgReducedTime

Table 2-3 Transformation of user-defined attributes.

In addition to the transformation steps described in the tables above, the following rules hold for the DBC transformation.

> For SignalGroups the SignalIPdu.ISignalToIPduMapping.TransferProperty is always Pending.

> The attribute CANBus.ILTxTimeout is not processed.

2.1.1 E2ETransfromer configuration via DBC

An E2ETransformer (E2EXF) may also be configured in the DBC file using the according user-defined attributes E2ETransformer described in Table 2-4. This will create one TransformationTechnology for every different E2EProfile in use, always in a TransformerChain together with the ComBasedTransformer.

Generically all different E2E Profiles are supported as far as they can be configured in the specified DBC attributes. Only the attributes present in the file are taken over to the created ARXML.

E2E parameters may be defined on message level: then only one SignalGroup inside the CAN message can be protected. If a SignalGroup is defined on the message, the E2E parameters will be used for protecting this SignalGroup, further Signals, not contained in the group, will be unprotected. If the message contains no SignalGroup the legacy converter creates one, containing all signals of the dedicated message.

The same E2E parameters may alternatively be specified on signal level as described in Table 2-4. Using this approach, the E2E parameters may protect several SignalGroups, if the signals containing the E2E parameters are associated to several groups.

Attribute	Autosar element
CANBus (<ProfileName> = P02) E2EP02ProfileName E2EP02ProfileBehavior E2EP02UpperHeaderBitsToShift E2EP02Offset E2EP02CrcOffset E2EP02CounterOffset E2EP02DataIdMode E2EP02DataIdNibbleOffset E2EP02SyncCounterInit E2EP02MaxNoNewOrRepeatedData E2EP02MaxDeltaCounter E2EP02MaxErrorStateInit E2EP02MaxErrorStateInvalid E2EP02MaxErrorStateValid E2EP02MinOkStateInit E2EP02MinOkStateInvalid E2EP02MinOkStateValid E2EP02WindowSize	DataTransformationSet.TransformationTechnology.EndToEndTransformationDescription ProfileName = CANBus.E2EP02ProfileName ProfileBehavior = CANBus.E2EP02ProfileBehavior UpperHeaderBitsToShift = CNABus.E2EP02UpperHeaderBitsToShift Offset = CANBus.E2EP02Offset CrcOffset = CANBus.E2EP02CrcOffset CounterOffset = CANBus.E2EP02CounterOffset E2EDataIdMode = CANBus.E2EP02DataIdMode DataIdNibbleOffset = CANBuas.E2EP02DataIdNibbleOffset SyncCounterInit = CANBus. E2EP02SyncCounterInit MaxNoNewOrRepeatedData = CANBus.E2EP02MaxNoNewOrRepeatedData MaxDeltaCounter = CANBus.E2EP02axDeltaCounter MaxErrorStateInit = CANBus.E2EP02MaxErrorStateInit MaxErrorStateInvalid = CANBus.E2EP02MaxErrorStateInvalid MaxErrorStateValid = CANBuas.E2EP02MaxErrorStateValid MinOkStateInit = CANBus.E2EP02MinOkStateInit MinOkStateInvalid = CANBus.E2EP02MinOkStateInvalid MinOkStateValid = CANBus.E2EP02MinOkStateValid WindowSize = CANBus.E2EP02WindowSize
CANFrame .E2EProfile .E2EDataId .E2EDataId01	ISignalGroup.End2EndTransformationISignalProps E2EProfile = CANFrame.E2EProfile E2EDataId = CANFrame.E2EDataId E2EDataId01 = CANFrame.E2EDataId01

.E2EDataId02	E2EDataId02 = CANFrame.E2EDataId02
.E2EDataId03	E2EDataId03 = CANFrame.E2EDataId03
.E2EDataId04	E2EDataId04 = CANFrame.E2EDataId04
.E2EDataId05	E2EDataId05 = CANFrame.E2EDataId05
.E2EDataId06	E2EDataId06 = CANFrame.E2EDataId06
.E2EDataId07	E2EDataId07 = CANFrame.E2EDataId07
.E2EDataId08	E2EDataId08 = CANFrame.E2EDataId08
.E2EDataId09	E2EDataId09 = CANFrame.E2EDataId09
.E2EDataId10	E2EDataId10 = CANFrame.E2EDataId10
.E2EDataId11	E2EDataId11 = CANFrame.E2EDataId11
.E2EDataId12	E2EDataId12 = CANFrame.E2EDataId12
.E2EDataId13	E2EDataId13 = CANFrame.E2EDataId13
.E2EDataId14	E2EDataId14 = CANFrame.E2EDataId14
.E2EDataId15	E2EDataId15 = CANFrame.E2EDataId15
.E2EDataId16	E2EDataId16 = CANFrame.E2EDataId16
.E2EDataLength	E2EDataLength = CANFrame.E2EDataLength
Signal	ISignalGroup.End2EndTransformationISignalProps
.E2EProfile	E2EProfile = Signal.E2EProfile
.E2EDataId	E2EDataId = Signal.E2EDataId
.E2EDataId01	E2EDataId01 = Signal.E2EDataId01
.E2EDataId02	E2EDataId02 = Signal.E2EDataId02
.E2EDataId03	E2EDataId03 = Signal.E2EDataId03
.E2EDataId04	E2EDataId04 = Signal.E2EDataId04
.E2EDataId05	E2EDataId05 = Signal.E2EDataId05
.E2EDataId06	E2EDataId06 = Signal.E2EDataId06
.E2EDataId07	E2EDataId07 = Signal.E2EDataId07
.E2EDataId08	E2EDataId08 = Signal.E2EDataId08
.E2EDataId09	E2EDataId09 = Signal.E2EDataId09
.E2EDataId10	E2EDataId10 = Signal.E2EDataId10
.E2EDataId11	E2EDataId11 = Signal.E2EDataId11
.E2EDataId12	E2EDataId12 = Signal.E2EDataId12
.E2EDataId13	E2EDataId13 = Signal.E2EDataId13
.E2EDataId14	E2EDataId14 = Signal.E2EDataId14
.E2EDataId15	E2EDataId15 = Signal.E2EDataId15
.E2EDataId16	E2EDataId16 = Signal.E2EDataId16
.E2EDataLength	E2EDataLength = Signal.E2EDataLength

Table 2-4 Transformation of user-defined attributes E2ETransformer.

2.1.2 Derivation towards Com instead of J1939

Derivation towards COM instead of J1939 may also be configured in the DBC file using the according user-defined attribute ‘ReRoutJ1939ToCom’ at node level. This will lead to the suppression of creation of J1939 components and redirects the J1939DcmIPdu to COM resulting in the creation of SignalIPdu replacing the J1939DcmIPdu.

Specification of node attribute ‘ReRoutJ1939ToCom’:

Objekttyp	Name	Wertetyp	Mini...	Max...	Default
Knoten	ReRoutJ1939ToCom	Enumeration	-	-	No

Sample usage of the node attribute ‘ReRoutJ1939ToCom’:

Name	Adresse	ReRoutJ1939ToCom
CAN	0x0	Yes
CAN 2	0xB3	Yes

2.2 LDF Transformation

The LDF file format does not provide any user-defined attributes or other OEM extensions. The subsequent table shows how LIN network objects are transformed in a generic way to AUTOSAR elements of the selected version as described in chapter 1.1.

LIN network object	AUTOSAR element
Signal	SystemSignal ShortName = Signal.Name Length = Signal.Bitcount BooleanType|IntegerType ShortName = "DT_" + Signal.Name LowerLimit = (Signal.Min-Signal.Offset)/Signal.Factor UpperLimit = (Signal.Max-Signal.Offset)/Signal.Factor CompuMethod ShortName = "CM_" + Signal.Name Unit ShortName = "U_" + Signal.Unit CompuInternalToPhys.CompuScale LowerLimit = Signal.TextualEncoding.LowerBound UpperLimit = Signal.TextualEncoding.UpperBound CompuConst = "Cx<Limit>_" + Signal.TextalEncoding.Text CompuInternalToPhys.CompuScale.CompuRationalCoeffs CompuNumerator = Signal.Offset, Signal.Factor ConstantSpecification ShortName = "C_" + Signal.Name BooleanLiteral|IntegerLiteral ShortName = "C_" + Signal.Name Value = Signal.LDFSignalSpecialValues
LINBus	LinCluster ShortName = LINBus.LINChannelPostfix ProtocolName = "LIN" ProtocolVersion = LINBus.ProtocolVersion Speed = LINBus.BaudRate PhysicalChannel ShortName = "CHNL"
LINBus .LINFrame(Unconditional)	Frame ShortName = LINFrame.Name + "__" + LinCluster.ShortName FrameLength = LINFrame.Size SignalIPdu|DcmIPdu|NPdu ShortName = LINFrame.Name + "__" + LinCluster.ShortName Length = 8*LINFrame.Size Frame.PduToFrameMapping ShortName = LINFrame.Name PackingByteOrder = LINBus.ByteOrder StartPosition = 0(Intel)|7(Motorola) PhysicalChannel.LinFrameTriggering ShortName = "FT_" + LINFrame.Name ChecksumType = LINFrame.CSModel Identifier = LINFrame.ID PhysicalChannel.IPduTriggering ShortName = "PT_" + LINFrame.Name
LINBus .LINFrame(Unconditional) .MappedSignal .Signal	ISignal ShortName = Signal.Name + "__" + LINFrame.Name + "__" + LinCluster.ShortName SignalIPdu.ISignalToIPduMapping ShortName = Signal.Name PackingByteOrder = MappedSignal.Intel|Motorola StartPosition = MappedSignal.Startbit PhysicalChannel.SignalTriggering ShortName = "ST_" + Signal.Name + "__" + LINFrame.Name
LINBus .LINFrame(Sporadic |EventTriggered)	SubstitutionFrame ShortName = LINFrame.Name + "__" + LinCluster.ShortName FrameLength = LINFrame.Size SubstitutionType = Sporadic|EventTriggered PhysicalChannel.LinFrameTriggering ShortName = "FT_" + LINFrame.Name ChecksumType = LINFrame.CSModel Identifier = LINFrame.ID

LINNode(Master|Slave)	EcuInstance ShortName = LINNode.Name EcuInstance.LinMaster ShortName = "CT_" + LinCluster.ShortName TimeBase = LINNode.Timebase TimeBaseJitter = LINNode.Jitter EcuInstance.LinSlave ShortName = "CT_" + LinCluster.ShortName ConfiguredNad = LINSlaveNode.ConfiguredNad LinErrorResponse = LINSlaveNode.ResponseErrorSignal ProtocolVersion = LINSlaveNode.ProtocolVersion EcuInstance.CommunicationConnector ShortName = "CN_" + LinCluster.ShortName EcuInstance.AssociatedIPduGroup(Rx) ShortName = LINNode.Name + "__" + LinCluster.ShortName + "_Rx" EcuInstance.AssociatedIPduGroup(Tx) ShortName = LINNode.Name + "__" + LinCluster.ShortName + "_Tx"
LINNode .RxLINFrame	CommunicationConnector.FramePort ShortName = "FP_" + RxLINFrame.Name + "_Rx" Direction = In CommunicationConnector.IPduPort ShortName = "PP_" + RxLINFrame.Name + "_Rx" Direction = In
LINNode .RxLINFrame .MappedSignal .Signal	CommunicationConnector.SignalPort ShortName = "SP_" + Signal.Name + "__" + RxLINFrame.Name + "_Rx" Direction = In
LINNode .TxLINFrame	CommunicationConnector.FramePort ShortName = "FP_" + TxLINFrame.Name + "_Tx" Direction = Out CommunicationConnector.IPduPort ShortName = "PP_" + TxLINFrame.Name + "_Tx" Direction = Out
LINNode .TxLINFrame .MappedSignal .Signal	CommunicationConnector.SignalPort ShortName = "SP_" + Signal.Name + "__" + TxLINFrame.Name + "_Tx" Direction = Out
LINBus .LINScheduleTable	LinCluster.LinScheduleTable ShortName = LINScheduleTable.Name Priority = 255 RunMode = RunContinuous
LINBus .LINScheduleTable .UnconditionalFrameSlot	LinFrameTriggering.RelativelyScheduledTiming Delay = UnconditionalFrameSlot.SlotDelay PositionInTable = UnconditionalFrameSlot.ID
LINBus .LINScheduleTable .DiagnosticFrameSlot	LinFrameTriggering.RelativelyScheduledTiming Delay = DiagnosticFrameSlot.SlotDelay PositionInTable = DiagnosticFrameSlot.ID
LINBus .LINScheduleTable .AssignFrameIdSlot	LinFrameTriggering.AssignFrameIdTiming Delay = AssignFrameIdSlot.SlotDelay PositionInTable = AssignFrameIdSlot.ID AssignedFrameTriggering ShortName = "FT_" + AssignFrameIdSlot.FrameToAssign.Name
LINBus .LINScheduleTable .UnassignFrameIdSlot	LinFrameTriggering.UnassignFrameIdTiming Delay = UnassignFrameIdSlot.SlotDelay PositionInTable = UnassignFrameIdSlot.ID UnassignedFrameTriggering ShortName = "FT_" + UnassignFrameIdSlot.FrameToUnassign.Name
LINBus .LINScheduleTable .AssignNADSlot	LinFrameTriggering.AssignNADTiming Delay = AssignNADSlot.SlotDelay PositionInTable = AssignNADSlot.ID NewNAD = AssignNADSlot.NewNAD
LINBus .LINScheduleTable .ConditionalChangeNADSlot	LinFrameTriggering.DataTiming Delay = ConditionalChangeNADSlot.SlotDelay PositionInTable = ConditionalChangeNADSlot.ID FreeFormatByteValues = ConditionalChangeNADSlot.DataBytes
LINBus	LinFrameTriggering.DataTiming

.LINScheduleTable .FreeFormatSlot	Delay = FreeFormatSlot.SlotDelay PositionInTable = FreeFormatSlot.ID FreeFormatByteValues = FreeFormatSlot.DataBytes
LINBus .LINScheduleTable .EventTriggeredFrameSlot	LinFrameTriggering.RelativelyScheduledTiming Delay = EventTriggeredFrameSlot.SlotDelay PositionInTable = EventTriggeredFrameSlot.ID
LINBus .LINScheduleTable .SporadicFrameSlot	LinFrameTriggering.RelativelyScheduledTiming Delay = SporadicFrameSlot.SlotDelay PositionInTable = SporadicFrameSlot.ID
LINBus .LINScheduleTable .DataDumpSlot	LinFrameTriggering.DataTiming Delay = DataDumpSlot.SlotDelay PositionInTable = DataDumpSlot.ID FreeFormatByteValues = DataDumpSlot.DataBytes
LINBus .LINScheduleTable .AssignFrameIdRangeSlot	LinFrameTriggering.DataTiming Delay = AssignFrameIdRangeSlot.SlotDelay PositionInTable = AssignFrameIdRangeSlot.ID FreeFormatByteValues = AssignFrameIdRangeSlot.DataBytes
LINBus .LINScheduleTable .SaveConfigurationSlot	LinFrameTriggering.DataTiming Delay = SaveConfigurationSlot.SlotDelay PositionInTable = SaveConfigurationSlot.ID FreeFormatByteValues = SaveConfigurationSlot.DataBytes

Table 2-5 Transformation of LIN network objects.

2.3 Fibex Transformation

The VLC supports the transformation of Fibex 2.0.1 files and of Fibex 3.0.0 or 3.1.0 files. The main difference between these Fibex versions is the modeling of PDUs. In Fibex 2.0.1 PDUs are modeled with signal groups, while in Fibex 3.0.0 and 3.1.0 PDUs are an explicit part of the XML schema. The table below shows how Fibex 2.0.1 elements are mapped to AUTOSAR elements of the selected version as described in chapter 1.1.

Fibex element	AUTOSAR element
PhysicalDimension	PhysicalDimesion ShortName = PhysicalDimension.ShortName LengthExp = PhysicalDimension.LengthExp MassExp = PhysicalDimension.MassExp TimeExp = PhysicalDimension.TimeExp CurrentExp = PhysicalDimension.CurrentExp TemperatureExp = PhysicalDimension.TemperatureExp MolarAmoutExp = PhysicalDimension.MolarAmoutExp LuminousIntensityExp = PhysicalDimension.LuminousIntensityExp
Unit	Unit ShortName = Unit.ShortName DisplayName = Unit.DisplayName FactorSiToUnit = Unit.FactorSiToUnit OffsetSiToUnit = Unit.OffsetSiToUnit
Coding	BooleanType|OpaqueType|IntegerType|RealType|CharType|StringType ShortName = Coding.ShortName LowerLimit = (Coding.CompuMethod.PhysConstr.LowerLimit - Coding.CompuMethod.CompuRationalCoeffs[0]) / Coding.CompuMethod.CompuRationalCoeffs[1] UpperLimit = (Coding.CompuMethod.PhysConstr.UpperLimit - Coding.CompuMethod.CompuRationalCoeffs[0]) / Coding.CompuMethod.CompuRationalCoeffs[1] InvalidValue.BooleanLiteral|OpaqueLiteral|IntegerLiteral |RealLiteral|CharLiteral|StringLiteral ShortName = Coding.ShortName Value = Coding.CompuMethod.InternalConstr.LowerLimit
Coding .CompuMethod	CompuMethod ShortName = "CM_" + CompuMethod.ShortName CompuInternalToPhys.CompuScale LowerLimit = CompuMethod.CompuScale.LowerLimit UpperLimit = CompuMethod.CompuScale.UpperLimit CompuConst = "Cx<Limit>_" + CompuMethod.CompuScale.CompuConst CompuRationalCoeffs = CompuMethod.CompuScale.CompuRationalCoeffs
Signal	SystemSignal ShortName = Signal.ShortName Length = Signal.Coding.BitLength ConstantSpecification ShortName = "C_" + Signal.ShortName BooleanLiteral|OpaqueLiteral|IntegerLiteral|RealLiteral |CharLiteral|StringLiteral ShortName = "C_" + Signal.ShortName Value = Signal.DefaultValue
Frame	Frame ShortName = Frame.ShortName FrameLength = Frame.ByteLength
SignalGroup	SignalIPdu|DcmIPdu|NmPdu|NPdu ShortName = SignalGroup.ShortName Length = SignalGroup.BitLength Frame.PduToFrameMapping ShortName = SignalGroup.ShortName PackingByteOrder = Intel StartPosition = Frame.SignalInstance.BitPosition - SignalGroup.OrderedSignal.BitPosition
Frame .SignalInstance .Signal	ISignal ShortName = Signal.ShortName + "__" + Frame.ShortName SignalIPdu.ISignalToIPduMapping ShortName = Signal.ShortName

	PackingByteOrder = SignalInstance.Intel|Motorola StartPosition = SignalInstance.BitPosition - Frame.PduToFrameMapping.StartPosition
Frame .Multiplexer .Switch	MultiplexedIPdu ShortName = Frame.ShortName Length = 8*Frame.ByteLength Frame.PduToFrameMapping ShortName = Frame.ShortName PackingByteOrder = Intel StartPosition = 0 MultiplexedIPdu.SelectorField ByteOrder = Switch.Intel|Motorola Length = Switch.BitLength StartPosition = Switch.BitPosition
Frame .Multiplexer .Data .SubFrame	MultiplexedIPdu.DynamicPart.DynamicPartAlternative SelectorFieldCode = SubFrame.SwitchCode SignalIPdu ShortName = SubFrame.ShortName + "_Mx" + SubFrame.SwitchCode Length = 8*Frame.ByteLength
Cluster(FlexRay)	FlexrayCluster ShortName = Cluster.ShortName MaxFrameLength = Cluster.MaxFrameLength ProtocolName = Cluster.Protocol ProtocolVersion = Cluster.ProtocolVersion Speed = Cluster.Speed ActionPointOffset = Cluster.ActionPointOffset Bit = Cluster.Bit/1000000 CasRxLowMin = Cluster.CasRxLowMin CasRxLowMax = Cluster.CasRxLowMax ColdStartAttempts = Cluster.ColdStartAttempts Cycle = Cluster.Cycle/1000000 DynamicSlotIdlePhase = Cluster.DynamicSlotIdlePhase ListenNoise = Cluster.ListenNoise MacroPerCycle = Cluster.MacroPerCycle MacrotickDuration = Cluster.Macrotick/1000000 MaxInitialisationError = Cluster.MaxInitializationError/100000 MaxWithoutClockCorrectionFatal = Cluster.MaxWithoutClockCorrectionFatal MaxWithoutClockCorrectionPassive = Cluster.MaxWithoutClockCorrectionPassive MinislotActionPointOffset = Cluster.MinislotActionPointOffset MinislotDuration = Cluster.Minislot NetworkIdleTime = Cluster.NIT NetworkManagementVectorLength = Cluster.NetworkManagementVectorLength NumberOfCycles = Cluster.NumberOfCycles NumberOfMinislots = Cluster.NumberOfMinislots NumberOfStaticSlots = Cluster.NumberOfStaticSlots OffsetCorrectionMax = Cluster.OffsetCorrectionMax/1000000 OffsetCorrectionStart = Cluster.OffsetCorrectionStart PayloadLengthStatic = Cluster.PayloadLengthStatic SampleClockPeriod = Cluster.SampleClockPeriod/1000000 StaticSlotDuration = Cluster.StaticSlot SymbolWindow = Cluster.SymbolWindow SyncFrameIdCountMax = Cluster.SyncNodeMax TransmissionStartSequenceDuration = Cluster.TSSTransmitter WakeupRxIdle = Cluster.WakeUpSymbolRxIdle WakeupRxLow = Cluster.WakeUpSymbolRxLow WakeupRxWindow = Cluster.WakeUpSymbolRxWindow WakeupTxActive = Cluster.WakeUpSymbolTxLow WakeupTxIdle = Cluster.WakeUpSymbolTxIdle
Channel	FlexrayCluster.FlexrayPhysicalChannel ShortName = Channel.ShortName ChannelName = Channel.FlexrayChannelName
Channel .FrameTriggering	FlexrayPhysicalChannel.FlexrayFrameTriggering ShortName = "FT_" + FrameTriggering.AbsolutelyScheduledTiming
Channel .FrameTriggering .AbsolutelyScheduledTiming	FlexrayFrameTriggering.AbsolutelyScheduledTiming SlotID = AbsolutelyScheduledTiming.SlotID BaseCycle = AbsolutelyScheduledTiming.BaseCycle CycleRepetition = AbsolutelyScheduledTiming.CycleRepetition

Channel .FrameTriggering .Frame .SignalGroup	FlexrayPhysicalChannel.IPduTriggering ShortName = "PT_" + SignalGroup.ShortName
Channel .FrameTriggering .Frame .SignalGroup .Signal	FlexrayPhysicalChannel.SignalTriggering ShortName = "ST_" + Signal.ShortName + "__"+ SignalGroup.ShortName
Channel .FrameTriggering .CyclicTiming	SignalIPdu.IPduTimingSpecification.CyclicTiming RepeatingTime = CyclicTiming.RepeatingTimeRange
Channel .FrameTriggering .EventControlledTiming	SignalIPdu.IPduTimingSpecification.EventControlledTiming RepetitionPeriod = EventControlledTiming.DebounceTimeRange
Channel .FrameTriggering .RequestControlledTiming	SignalIPdu.IPduTimingSpecification.RequestControlledTiming ResponseTime = RequestControlledTiming.ResponseTimeRange
Ecu	EcuInstance ShortName = Ecu.ShortName
Ecu .Controller	EcuInstance.FlexrayCommunicationController ShortName = Controller.ShortName AcceptedStartupRange = Controller.AcceptedStartupRange AllowHaltDueToClock = Controller.AllowHaltDueToClock AllowPassiveToActive = Controller.AllowPassiveToActive ClusterDriftDamping = Controller.ClusterDriftDamping DecodingCorrection = Controller.DecodingCorrection DelayCompensationA = Controller.DelayCompensationA DelayCompensationB = Controller.DelayCompensationB ExternOffsetCorrection = Controller.ExternOffsetCorrection ExternRateCorrection = Controller.ExternRateCorrection KeySlotId = Controller.KeySlotUsage.StartupSync | Controller.KeySlotUsage.Sync KeySlotUsedForStartUp = Controller.KeySlotUsage.StartupSync!=null KeySlotUsedForSync = Controller.KeySlotUsage.StartupSync!=null | Controller.KeySlotUsage.Sync !=null LatestTx = Controller.LatestTx ListenTimeout = Controller.ListenTimeout MacroInitialOffsetA = Controller.MacroInitialOffsetA MacroInitialOffsetB = Controller.MacroInitialOffsetB MaximumDynamicPayloadLength = Controller.MaxDynamicPayloadLength MicroInitialOffsetA = Controller.MicroInitialOffsetA MicroInitialOffsetB = Controller.MicroInitialOffsetB MicroPerCycle = Controller.MicroPerCycle MicrotickDuration = Controller.Microtick/1000000 OffsetCorrectionOut = Controller.OffsetCorrectionOut RateCorrectionOut = Controller.RateCorrectionOut SamplesPerMicrotick = Controller.SamplesPerMicrotick WakeUpPattern = Controller.WakeUpPattern
Ecu .Connector	EcuInstance.FlexRayCommunicationConnector ShortName = "CN_" + Cluster.ShortName + "__" + Connector.Channel.ShortName TpAddress = Ecu.DiagnosticAddress[Physical] WakeUpChannel = Connector.WakeUpChannel EcuInstance.AssociatedIPduGroup(Rx) ShortName = Ecu.ShortName + "__" + Connector.Channel.ShortName + "_Rx" EcuInstance.AssociatedIPduGroup(Tx) ShortName = Ecu.ShortName + "__" + Connector.Channel.ShortName + "_Tx"
Ecu .Connector .InputPort .FrameTriggering	FlexRayCommunicationConnector.FramePort ShortName = "FP_" + FrameTriggering.AbsolutelyScheduledTiming + "_Rx" Direction = In
Ecu .Connector .InputPort .SignalInstance .Signal .SignalGroup	FlexRayCommunicationConnector.IPduPort ShortName = "PP_" + SignalGroup.ShortName + "_Rx" Direction = In FlexRayCommunicationConnector.SignalPort ShortName = "SP_" + Signal.ShortName + "__" + SignalGroup.ShortName + "_Rx" Direction = In

Ecu .Connector .OutputPort .FrameTriggering	FlexRayCommunicationConnector.FramePort ShortName = "FP_" + FrameTriggering.AbsolutelyScheduledTiming + "_Tx" Direction = Out
Ecu .Connector .OutputPort .SignalInstance .Signal .SignalGroup	FlexRayCommunicationConnector.IPduPort ShortName = "PP_" + SignalGroup.ShortName + "_Tx" Direction = Out FlexRayCommunicationConnector.SignalPort ShortName = "SP_" + Signal.ShortName + "__" + SignalGroup.ShortName + "_Tx" Direction = Out

Table 2-6 Transformation of Fibex 2.0.1 elements.

The subsequent table shows how Fibex 3.0.0 or 3.1.0 elements are mapped to AUTOSAR elements of the selected version as described in chapter 1.1.

Fibex element	AUTOSAR Element
PhysicalDimension	see Table 2-6
Unit	see Table 2-6
Coding	see Table 2-6
Signal	see Table 2-6
Frame	see Table 2-6
Frame .PduInstance	Frame.PduToFrameMapping ShortName = PduInstance.Pdu.ShortName PackingByteOrder = PduInstance.Intel|Motorola StartPosition = PduInstance.BitPosition UpdateIndicationBitPosition = PduInstance.PduUpdateBitPosition
Pdu	SignalIPdu|MultiplexedIPdu|DcmIPdu|NmPdu|NPdu ShortName = Pdu.ShortName Length = 8*Pdu.ByteLength
Pdu .SignalInstance .Signal	ISignal ShortName = Signal.ShortName + "__" + Pdu.ShortName SignalIPdu.ISignalToIPduMapping ShortName = Signal.ShortName PackingByteOrder = SignalInstance.Intel|Motorola StartPosition = SignalInstance.BitPosition
Pdu .Multiplexer .Switch	MultiplexedIPdu.SelectorField ByteOrder = Switch.Intel|Motorola Length = Switch.BitLength StartPosition = Switch.BitPosition
Pdu .Multiplexer .DynamicPart .SwitchedPduInstance	MultiplexedIPdu.DynamicPart.DynamicPartAlternative SelectorFieldCode = SwitchedPduInstance.SwitchCode SignalIPdu ShortName = SwitchedPduInstance.Pdu.ShortName Length = 8*SwitchedPduInstance.Pdu.ByteLength
Cluster(FlexRay)	see Table 2-6
Channel	see Table 2-6
Channel .FrameTriggering	see Table 2-6
Channel .FrameTriggering .AbsolutelyScheduledTiming	see Table 2-6
Channel .PduTriggering	FlexrayPhysicalChannel.IPduTriggering ShortName = "PT_" + PduTriggering.Pdu.ShortName
Channel .PduTriggering .Pdu .SignalInstance .Signal	FlexrayPhysicalChannel.SignalTriggering ShortName = "ST_" + Signal.ShortName + "__" + PduTriggering.Pdu.ShortName
Channel .PduTriggering .CyclicTiming	SignalIPdu.IPduTimingSpecification.CyclicTiming FinalRepetitions = CyclicTiming.FinalRepetitions RepeatingTime = CyclicTiming.RepeatingTimeRange StartingTime = CyclicTiming.StartingTimeRange
Channel .PduTriggering .EventControlledTiming	SignalIPdu.IPduTimingSpecification.EventControlledTiming NumberOfRepeats = EventControlledTiming.FinalRepetitions RepetitionPeriod = EventControlledTiming.DebounceTimeRange
Channel .PduTriggering .RequestControlledTiming	SignalIPdu.IPduTimingSpecification.RequestControlledTiming ResponseTime = RequestControlledTiming.ResponseTimeRange
Ecu	see Table 2-6
Ecu .Controller	see Table 2-6
Ecu .Connector	see Table 2-6
Ecu .Connector .InputPort .FrameTriggering	see Table 2-6

Ecu .Connector .InputPort .IncludedPdu .PduTriggering	FlexRayCommunicationConnector.IPduPort ShortName = "PP_" + PduTriggering.Pdu.ShortName + "_Rx" Direction = In
Ecu .Connector .InputPort .IncludedPdu .IncludedSignal .SignalInstance	FlexRayCommunicationConnector.SignalPort ShortName = "SP_" + SignalInstance.Signal.ShortName + "__" + IncludedPdu.PduTriggering.Pdu.ShortName + "_Rx" Direction = In
Ecu .Connector .OutputPort .FrameTriggering	see Table 2-6
Ecu .Connector .OutputPort .IncludedPdu .PduTriggering	FlexRayCommunicationConnector.IPduPort ShortName = "PP_" + PduTriggering.Pdu.ShortName + "_Tx" Direction = Out
Ecu .Connector .OutputPort .IncludedPdu .IncludedSignal .SignalInstance	FlexRayCommunicationConnector.SignalPort ShortName = "SP_" + SignalInstance.Signal.ShortName + "__" + IncludedPdu.PduTriggering.Pdu.ShortName + "_Tx" Direction = Out
TpConfig .TpAddress	FlexrayPhysicalChannel.TpAddress ShortName = "TA_" + TpAddress TpAddress = TpAddress
TpConfig .TpChannel	FlexrayPhysicalChannel.FlexrayTpChannel AckType = TpChannel.AckType ExtendedAddressing = TpChannel.AddressingType == FrtpTb MaxBs = TpChannel.MaxBlockSize MaxRetries = TpChannel.MaxRetries MaximumMessageLength = TpChannel.MaximumMessageLength MulticastSegmentation = TpChannel.GroupSegmentation TimeoutBs = TpChannel.TimeoutBs TimeoutCr = TpChannel.TimeoutCr TransmitCancellation = TpChannel.TransmitCancellation
TpConfig .TpChannel .TpConnection	FlexrayTpChannel.FlexRayTpConnection FlexrayTpChannel.FlexRayTpConnection.DirectTpSdu ShortName = TpConnection.ShortName + "_Rq" FlexrayTpChannel.FlexRayTpConnection.ReversedTpSdu ShortName = TpConnection.ShortName + "_Rs"
TpConfig .TpNode	FlexrayPhysicalChannel.FlexrayTpNode ShortName = TpNode.ShortName FlexrayPhysicalChannel.FlexrayTpChannel MaxAr = TpNode.MaxAr MaxAs = TpNode.MaxAs MaxBufferRequest = TpNode.BufferRequest MaxFrIf = TpNode.MaxFrif MinimumSeparationTime = TpNode.Stmin TimeBuffer = TpNode.TimeBuffer TimeFrIf = TpNode.TimeFrif TimeoutAr = TpNode.TimeoutAr TimeoutAs = TpNode.TimeoutAs

Table 2-7 Transformation of Fibex 3.0.0/3.1.0 elements.

2.4 Extension File

The Vector System Description Extension (VSDE) file is used to supplement the content of DBC-, LDF- or Fibex files. An extension file defines certain communication elements which might be missing in the original legacy communication databases, or which cannot be specified with these communication databases. The table below explains the extension elements which are supported so far.

VSDE element	Description
<FILTERING> <FLEXRAY-CLUSTER-REF CHANNEL="A">FlexRay01 </FLEXRAY-CLUSTER-REF> </FILTERING>	The dual channel FlexRayCluster FlexRay01 is filtered for its A channel. This feature is supported for Fibex databases.
<CAN-CLUSTER-NAME> <CAN-CLUSTER-REF>Can01</CAN-CLUSTER-REF> <SHORT-NAME>Can01NewName</SHORT-NAME> </CAN-CLUSTER-NAME>	The CanCluster Can01 obtains a new name Can01NewName. Similarly, LinClusters and FlexrayClusters can be renamed. This feature is supported for DBC, LDF and Fibex databases.
<ECU-INSTANCE-NAME> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SHORT-NAME>Ecu01NewName</SHORT-NAME> </ECU-INSTANCE-NAME>	The ECU Ecu01 obtains a new name Ecu01NewName. This feature is supported for DBC, LDF and Fibex databases.
<SYSTEM-SIGNAL-NAME> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> <SHORT-NAME>Sig01NewName</SHORT-NAME> </SYSTEM-SIGNAL-NAME>	The signal Sig01 within pdu Pdu01 obtains a new name Sig01NewName. Signal renaming is used, e.g., to distinguish signals of the same name in different pdus. This feature is supported for DBC, LDF and Fibex databases.
<SYSTEM-SIGNAL-GROUP> <SHORT-NAME>SG_SigGrp01</SHORT-NAME> <SYSTEM-SIGNAL-REFS> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> <SYSTEM-SIGNAL-REF>Sig02</SYSTEM-SIGNAL-REF> </SYSTEM-SIGNAL-REFS> </SYSTEM-SIGNAL-GROUP>	The signals Sig01 and Sig02 are aggregated to a new signal group SG_SigGrp01. The signals must be defined in the same database. Each pdu must contain all or none of these signals. This feature is supported for DBC, LDF and Fibex databases.
<SAFETY-PDU> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> <CREATE-PDU-GAP-SIGNALS>true</CREATE-PDU-GAP-SIGNALS> </SAFETY-PDU>	All signals of pdu Pdu01 are aggregated to a new signal group SG_Pdu01. Optionally, the pdu gaps are filled with artificial gap signals, when then also become part of the new signal group. This feature is supported for DBC, LDF and Fibex databases.
<BIDIRECTIONAL-PDU> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> </BIDIRECTIONAL-PDU>	The pdu Pdu01 can be send and received by the same ECU. This feature is supported for DBC and Fibex databases.
<PDU-GROUP> <SHORT-NAME>PduGrp01</SHORT-NAME> <SIGNAL-I-PDU-REFS> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> <SIGNAL-I-PDU-REF>Pdu02</SIGNAL-I-PDU-REF> </SIGNAL-I-PDU-REFS> </PDU-GROUP>	For each cluster and ECU which sends or receives the pdus Pdu01 or Pdu02 a new Tx or Rx pdu group is created. The pdus are not assigned to the standard pdu groups created by the VLC. This feature is supported for DBC, LDF and Fibex databases.
<PDU-GROUP-DEFINITION> <FLEXRAY-CLUSTER-REF>FlexRay01</FLEXRAY-CLUSTER-REF> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <CHANNEL-SPECIFIC>true</CHANNEL-SPECIFIC> </PDU-GROUP-DEFINITION>	The dual channel FlexRayCluster FlexRay01 obtains for its ECU Ecu01 two standard Tx pdu groups and two standard Rx pdu groups – one for each channel and direction, respectively. This feature is supported for Fibex databases.

<SIGNAL-UPDATE-DEFINITION> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> <UPDATE-INDICATION-SIGNAL-REF> SigUpd01_UB</UPDATE-INDICATION-SIGNAL-REF> <UPDATED-SIGNALS> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> </UPDATED-SIGNALS> <UPDATED-SIGNAL-GROUPS> <SYSTEM-SIGNAL-GROUP-REF> SG_SigGrp01</SYSTEM-SIGNAL-GROUP-REF> </UPDATED-SIGNAL-GROUPS> </SIGNAL-UPDATE-DEFINITION>	The signal SigUpd01_UB within pdu Pdu01 serves as update signal for the signal Sig01 and the signal group SG_SigGrp01. The update signal can be used for one or more signals and signal groups within a pdu at the same time. This feature is supported for DBC, LDF and Fibex databases.
<I-PDU-TIMING> <SIGNAL-I-PDU-REF>Pdu01</SIGNAL-I-PDU-REF> <NUMBER-OF-REPETITIONS>10</NUMBER-OF-REPETITIONS> <REPETITION-PERIOD>0.001</REPETITION-PERIOD> <MINIMUM-DELAY>0.01</MINIMUM-DELAY> <SIGNAL-TIMINGS> <SIGNAL-TIMING> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> <SIGNAL-SEND-TYPE>ON-CHANGE</SIGNAL-SEND-TYPE> </SIGNAL-TIMING> </SIGNAL-TIMINGS> <ACCESS-RIGHTS>READ-ONLY</ACCESS-RIGHTS> </I-PDU-TIMING>	The timing elements NumberOfRepetitions, RepetitionPeriod and MinimumDelay of pdu Pdu01 override the corresponding database settings. Further, the SignalSendType timing element of signal Sig01 overrides the signal specific timing settings. Finally, the element AccessRights defines whether the timing data later can be changed in the Vector tool chain. This feature is supported for DBC databases. The AccessRights element is also supported for LDF and Fibex databases.
<CAN-TP-CONNECTION> <SHORT-NAME>Can01_Pdu01_Pdu02</SHORT-NAME> <CAN-CLUSTER-REF>Can01</CAN-CLUSTER-REF> <DATA-PDU-REF>Pdu01</DATA-PDU-REF> <FLOW-CONTROL-PDU-REF>Pdu02</FLOW-CONTROL-PDU-REF> </CAN-TP-CONNECTION>	The directly opposed pdus Pdu01 and Pdu02 of CanCluster Can01 are combined to a new CanTpConnection Can01_Pdu01_Pdu02. The VSDE internal CanTpConnection name can be referred by TpHighLevelRoutings. Similarly, pdus can be combined to a LinTpConnection. This feature is supported for DBC and LDF databases.
<PDUR-MESSAGE-ROUTING> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SOURCE-CAN-CLUSTER-REF>Can01</SOURCE-CAN-CLUSTER-REF> <TARGET-CAN-CLUSTER-REF>Can02</TARGET-CAN-CLUSTER-REF> <I-PDU-MAPPINGS> <I-PDU-MAPPING> <ROUTE-DLC>true</ROUTE-DLC> <SOURCE-I-PDU-REF>Pdu01</SOURCE-I-PDU-REF> <SOURCE-SIGNALS> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> </SOURCE-SIGNALS> <TARGET-I-PDU-REF>Pdu02</TARGET-I-PDU-REF> </I-PDU-MAPPING> </I-PDU-MAPPINGS> </PDUR-MESSAGE-ROUTING>	The pdu Pdu01 of CanCluster Can01 is routed via the gateway ECU Ecu01 to the pdu Pdu02 of CanCluster Can02. The pdu Pdu01 will be routed by the PDUR module, and also its DLC value will be routed. The signal Sig01 of pdu Pdu01 is received by the gateway ECU Ecu01, all other signals of pdu Pdu01 are not received by Ecu01. This feature is supported for DBC, LDF and Fibex databases.
<COM-MESSAGE-ROUTING> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SOURCE-CAN-CLUSTER-REF>Can01</SOURCE-CAN-CLUSTER-REF> <TARGET-CAN-CLUSTER-REF>Can02</TARGET-CAN-CLUSTER-REF> <I-PDU-MAPPINGS> <I-PDU-MAPPING> <PROCESSING>IMMEDIATE</PROCESSING> <ROUTE-DLC>true</ROUTE-DLC> <SOURCE-I-PDU-REF>Pdu01</SOURCE-I-PDU-REF> <SOURCE-SIGNALS> <SYSTEM-SIGNAL-REF>Sig01</SYSTEM-SIGNAL-REF> </SOURCE-SIGNALS> <SOURCE-EXCLUDE-SIGNALS> <SYSTEM-SIGNAL-REF>Sig02</SYSTEM-SIGNAL-REF> </SOURCE-EXCLUDE-SIGNALS> <TARGET-I-PDU-REF>Pdu02</TARGET-I-PDU-REF> <TARGET-EXCLUDE-SIGNALS> <SYSTEM-SIGNAL-REF>Sig02</SYSTEM-SIGNAL-REF> </TARGET-EXCLUDE-SIGNALS> </I-PDU-MAPPING> </I-PDU-MAPPINGS>	The pdu Pdu01 of CanCluster Can01 is routed via the gateway ECU Ecu01 to the pdu Pdu02 of CanCluster Can02. The pdu Pdu01 will be routed immediately by the COM module, and also its DLC value will be routed. The signal Sig01 of pdu Pdu01 is received by the gateway ECU Ecu01, all other signals of pdu Pdu01 are not received by Ecu01. The signal Sig02 of pdu Pdu01 is excluded from the routings merge algorithm for the COM module. This avoids conflicts with an OnChange sending behavior of COM routed signals. This feature is supported for DBC, LDF and Fibex databases.

</COM-MESSAGE-ROUTING>	
<COM-SIGNAL-ROUTING> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SOURCE-CAN-CLUSTER-REF>Can01</SOURCE-CAN-CLUSTER-REF> <TARGET-CAN-CLUSTER-REF>Can02</TARGET-CAN-CLUSTER-REF> <SIGNAL-MAPPINGS> <SIGNAL-MAPPING> <PROCESSING>DEFERED</PROCESSING> <SOURCE-I-PDU-REF>Pdu01</SOURCE-I-PDU-REF> <SOURCE-SIGNAL-REF>Sig01</SOURCE-SIGNAL-REF> <TARGET-I-PDU-REF>Pdu02</TARGET-I-PDU-REF> <TARGET-SIGNAL-REF>Sig02</TARGET-SIGNAL-REF> </SIGNAL-MAPPING> </SIGNAL-MAPPINGS> </COM-SIGNAL-ROUTING>	The signal Sig01 within pdu Pdu01 of CanCluster Can01 is routed via the gateway ECU Ecu01 to the signal Sig02 within pdu Pdu02 of CanCluster Can02. The signal Sig01 will be routed deferred by the COM module. This feature is supported for DBC, LDF and Fibex databases.
<TP-HIGH-LEVEL-ROUTING> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SOURCE-CAN-TP-CONNECTION-REF>Can01_Pdu01_Pdu02 </SOURCE-CAN-TP-CONNECTION-REF> <TARGET-CAN-TP-CONNECTION-REF>Can02_Pdu03_Pdu04 </TARGET-CAN-TP-CONNECTION-REF> </TP-HIGH-LEVEL-ROUTING>	The CanTpConnection Can01_Pdu01_Pdu02 is routed via the gateway ECU Ecu01 to the CanTpConnection Can02_Pdu03_Pdu04. CanTpConnections and LinTpConnections are defined by the VSDE file, while FlexrayTpConnections are provided by Fibex databases. This feature is supported for DBC, LDF and Fibex databases.
<TP-LOW-LEVEL-ROUTING> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <SOURCE-CAN-CLUSTER-REF>Can01</SOURCE-CAN-CLUSTER-REF> <TARGET-CAN-CLUSTER-REF>Can02</TARGET-CAN-CLUSTER-REF> <N-PDU-MAPPINGS> <N-PDU-MAPPING> <SOURCE-N-PDU-REF>Pdu01</SOURCE-N-PDU-REF> <TARGET-N-PDU-REF>Pdu02</TARGET-N-PDU-REF> </N-PDU-MAPPING> </N-PDU-MAPPINGS> </TP-LOW-LEVEL-ROUTING>	The n-pdu Pdu01 of CanCluster Can01 is routed via the gateway ECU Ecu01 to the n-pdu Pdu02 of CanCluster Can02. This feature is supported for DBC databases.
<PNC-CONFIGURATION> <PNC-VECTOR-LENGTH>3</PNC-VECTOR-LENGTH> <PNC-VECTOR-OFFSET>5</PNC-VECTOR-OFFSET> <PNC-CLUSTERS> <PNC-CLUSTER> <CAN-CLUSTER-REF>Can01</CAN-CLUSTER-REF> <PNC-ECUS> <PNC-ECU> <ECU-INSTANCE-REF>Ecu01</ECU-INSTANCE-REF> <PNC-GATEWAY-TYPE>ACTIVE</PNC-GATEWAY-TYPE> <PNC-WAKEUP-CAN-ID>452984832 </PNC-WAKEUP-CAN-ID> <PNC-WAKEUP-CAN-ID-EXTENDED>true </PNC-WAKEUP-CAN-ID-EXTENDED> <PNC-WAKEUP-CAN-ID-MASK>127 </PNC-WAKEUP-CAN-ID-MASK> <PNC-WAKEUP-DATA-MASK>4611686018427387904 </PNC-WAKEUP-DATA-MASK> <PNC-WAKEUP-DLC>8</PNC-WAKEUP-DLC> <PNC-GROUPS> <PNC-GROUP> <PNC-IDENTIFIER>1</PNC-IDENTIFIER> <COMMUNICATION-DIRECTION>IN <COMMUNICATION-DIRECTION> <SIGNAL-I-PDU-REFS> <SIGNAL-I-PDU-REF>Pdu01 </SIGNAL-I-PDU-REF> </SIGNAL-I-PDU-REFS> <MULTIPLEXED-I-PDU-REFS> <MULTIPLEXED-I-PDU-REF>Pdu02 </MULTIPLEXED-I-PDU-REF> </MULTIPLEXED-I-PDU-REFS> </PNC-GROUP> </PNC-GROUPS> </PNC-ECU> </PNC-ECUS>	The pdus Pdu01 and Pdu02 of CanCluster Can01 are combined to a PNC group for ECU Ecu01 and the partial network with the ID 1. A partial network is defined by all PNC groups which refer the same partial network ID. This feature is supported for DBC and Fibex databases.

</PNC-CLUSTER> </PNC-CLUSTERS> </PNC-CONFIGURATION>	
<SYSTEM-SIGNALS> <SYSTEM-SIGNAL> <SHORT-NAME>Signal_MasterClockSignal</SHORT-NAME> <ARRAY-TYPE>DYNAMIC-ARRAY</ARRAY-TYPE> </SYSTEM-SIGNAL> </SYSTEM-SIGNALS>	Signal_MasterClockSignal and the associated datatype are generated as a dynamic array (UINT8_DYN). Other possible values are STATIC-ARRAY (static array, UINT8_N) and NO-ARRAY (no array, UINT64). This VSDE transformation is only possible for signals with a length greater 32 bit. Additionally, NO-ARRAY can only be used for signals up to 64 bit. Note that for STATIC-ARRAY and DYNAMIC-ARRAY the corresponding signals length (in bit) must be a multiple of 8 and the signal must start at a byte boarder. This feature is supported for DBC and Fibex databases.
<E2E-CONFIGURATION> <LIN-CLUSTER-REF>LIN00</LIN-CLUSTER-REF> <E2E-NETWORK-CONFIGURATION> <E2E-PROFILE>P01</E2E-PROFILE> <E2E-PROFILE-NAME>PROFILE_01</E2E-PROFILE-NAME> <E2E-UPPER-HEADER-BITS-TO-SHIFT>0</E2E-UPPER-HEADER-BITS-TO-SHIFT> <E2E-MAX-DELTA-COUNTER>1</E2E-MAX-DELTA-COUNTER> <E2E-MAX-ERROR-STATE-INIT>15</E2E-MAX-ERROR-STATE-INIT> <E2E-MAX-ERROR-STATE-INVALID>15</E2E-MAX-ERROR-STATE-INVALID> <E2E-MAX-ERROR-STATE-VALID>15</E2E-MAX-ERROR-STATE-VALID> <E2E-MIN-OK-STATE-INIT>1</E2E-MIN-OK-STATE-INIT> <E2E-MIN-OK-STATE-INVALID>1</E2E-MIN-OK-STATE-INVALID> <E2E-MIN-OK-STATE-VALID>1</E2E-MIN-OK-STATE-VALID> <E2E-WINDOW-SIZE>15</E2E-WINDOW-SIZE> </E2E-NETWORK-CONFIGURATION> <E2E-FRAME-CONFIGURATION> <E2E-FRAME-NAME>LIN_E2E_Transmit</E2E-FRAME-NAME> <E2E-CLUSTER-NAME>LIN7</E2E-CLUSTER-NAME> <E2E-SIGNAL-GROUP-CONFIGURATION> <E2E-SIGNAL-GROUP-NAME>LIN_E2E_Transmit_SigGroup_A</E2E-SIGNAL-GROUP-NAME> <E2E-PROFILE>P01</E2E-PROFILE> </E2E-SIGNAL-GROUP-CONFIGURATION> </E2E-FRAME-CONFIGURATION> </E2E-CONFIGURATION>	E2E protection specified for signal group, <E2E-SIGNAL-GROUP-NAME> LIN_E2E_Transmit_SigGroup_A, which must be defined as <SYSTEM-SIGNAL-GROUP> referring to the group signals contained. To clearly identify the correct frame, the associated cluster is referenced via <LIN-CLUSTER-REF>. The E2E protection profile is referred to in <E2E-PROFILE> matching the <E2E-NETWORK-CONFIGURATION> / <E2E-PROFILE>. Example specifies only mandatory attributes, for full list of attributes see XML schema ExtractExtension.xsd.

Table 2-8 Vector System Description Extension file elements.

The extension file is provided as a file parameter to the VLC. Its XML schema is described with the ExtractExtension.xsd file.

3 Glossary and Abbreviations

3.1 Glossary

Term	Description

3.2 Abbreviations

Abbreviation	Description
DPA	DaVinci Project Assistant
VLC	Vector Legacy Converter
VSDE	Vector System Description Extension

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